VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Mission Agnostic Service Infrastructure (MASI)

AFLCMC/WA

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1.0 Overview

Mission Agnostic Service Infrastructure (MASI) is the set of Services responsible for tasks such as startup, health assessments, deployment, and logging. These Services support the operation of Skills.

1.1 Purpose

The MASI Volume describes the functionality of the Agile Mission Suite (AMS) Government Reference Architecture (GRA) MASI elements.

1.2 Scope

The MASI Volume aids a Mission System Integrator (MSI) in designing and implementing the functions of the MASI elements.

1.3 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or Minimum Procurable Unit (MPU) provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.3.1** for a list of informative references for the MASI Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Software Architecture Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	Cyber IDD	14.0
OMS Standard	OMS Standard	Version identified by the MSI

Table 1.3.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About MASI and Related Entities

2.0 Context

Figure 2.0.1 depicts the relationship between MASI, Skills, and Services. MASI is part of an AMS GRA mission system and is responsible for the deployment of container-based Services, including AMS GRA Services and Open Mission Systems (OMS) Services. A MASI implementation may also contain AMS GRA Services and OMS Service MPUs to provide mission-agnostic functionality. All containerized services within an AMS GRA mission system have a dependency on MASI for container management.

Once MASI is started, MASI controls and monitors operation of Services within containers using the container framework.

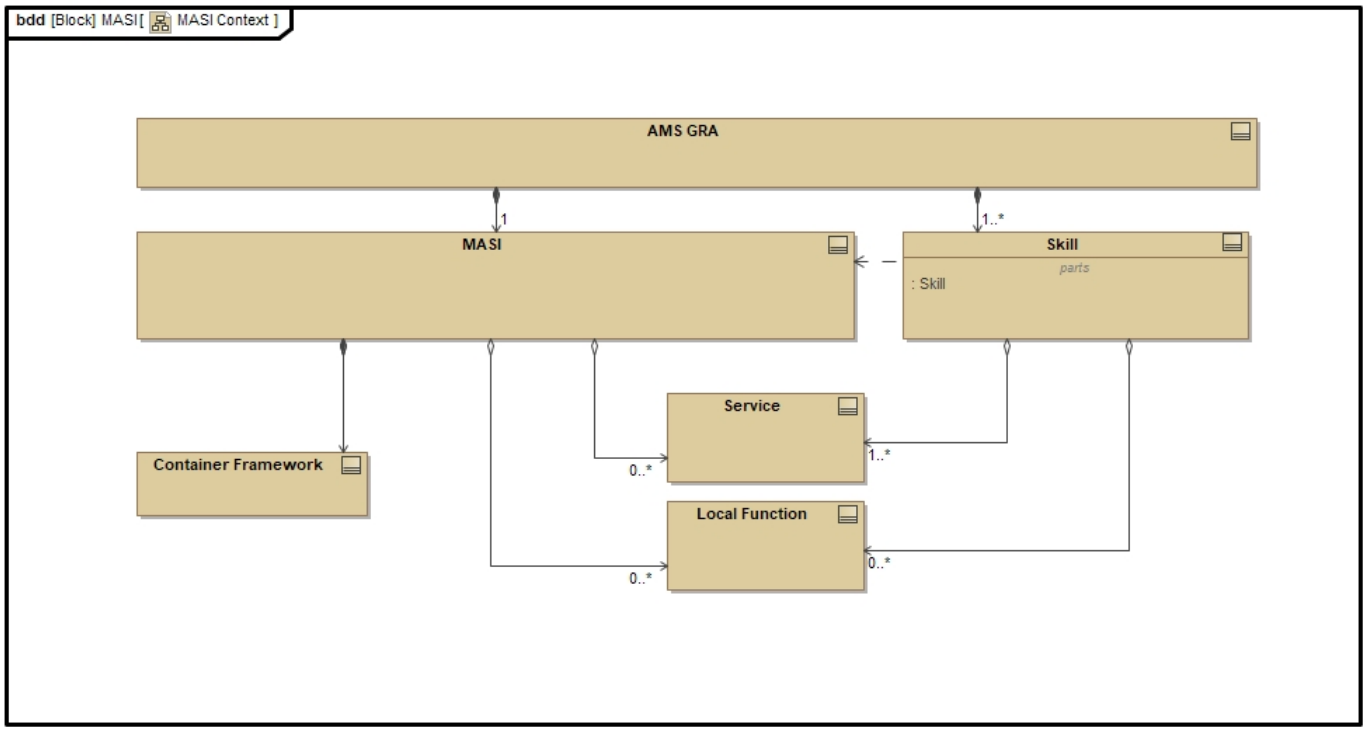


Figure 2.0.1: The MASI Context Diagram Depicts the MASI Structure and the Relationship Between MASI, Skill, and Services.

3.0 Mission Agnostic Service Infrastructure (MASI)

The MSI is responsible for providing a MASI implementation.[1]

3.1 MASI Functions

A MASI implements behaviors, shown in **Figure 3.1.1**, relative to the AMS GRA processing hardware and software services. The MASI functionality is allocated to the following categories: Manage System Faults, Facilitate Networking Connectivity, Provide Local Function Information, Monitor Health and Status, Monitor Computing Resources, Support Services Versioning and Configuration, and Manage Service Lifecycles.

[1] This section omits specific implementation details. For an example MASI implementation, see the Starter Kit (SK).

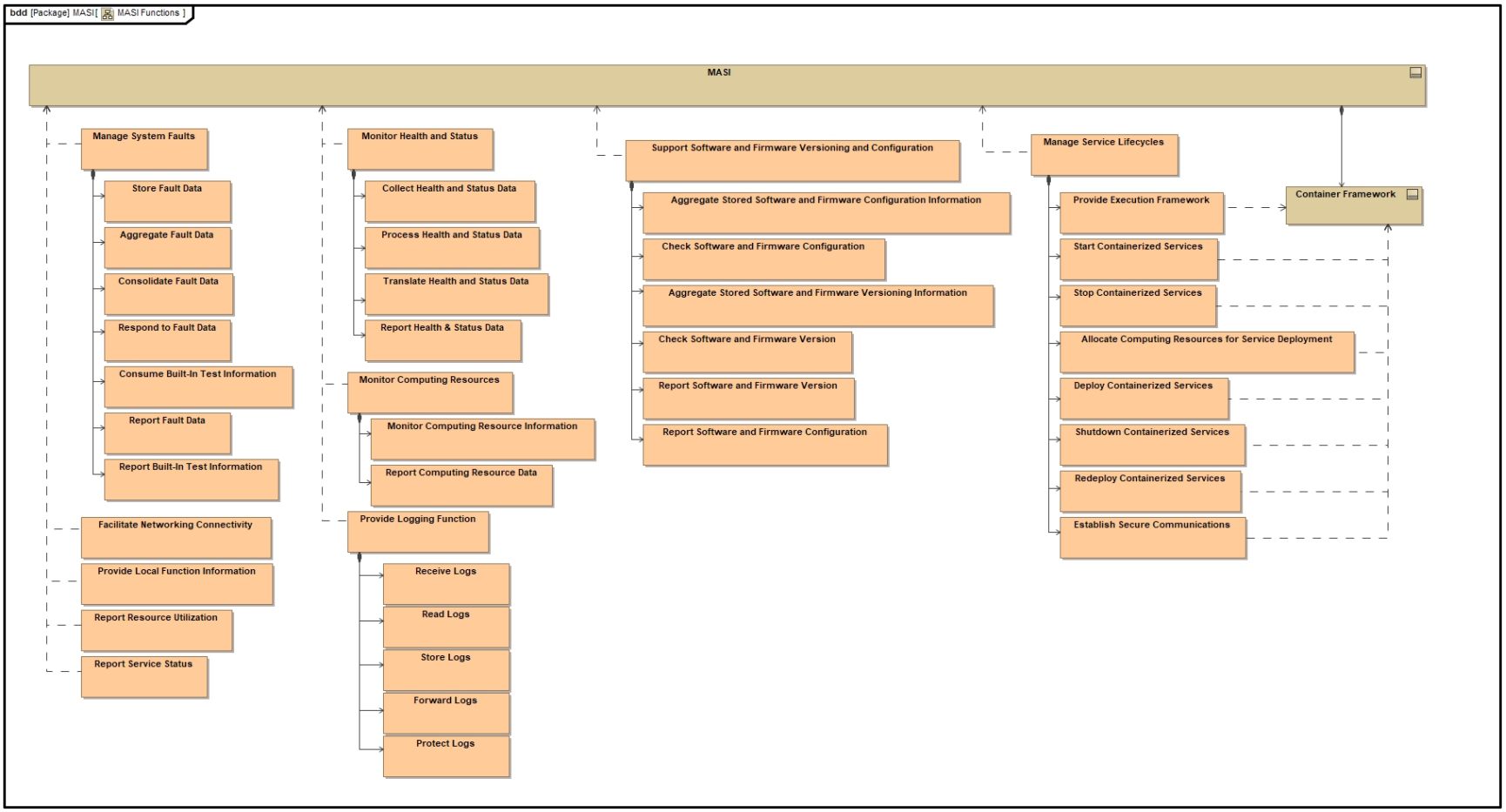


Figure 3.1.1: The MASI Functional Decomposition Diagram Shows Behaviors Implemented by MASI.

3.1.1 Required Functions

Table 3.1.1 describes the minimum functionality for a compliant MASI implementation.

Name	Documentation
Aggregate Fault Data	This function aggregates fault data.
Aggregate Stored Software and Firmware Configuration Information	This function enables the system to gather and process multiple Software and Firmware configurations stored on the Mission System.
Aggregate Stored Software and Firmware Versioning Information	This function enables the system to gather and process multiple versions of Software and Firmware stored on the Mission System.
Allocate Computing Resources for Service Deployment	This function allocates system resources for deployment of services based on the deployment and recovery strategy.
Check Software and Firmware Configuration	This function enables the system to reference Software and Firmware configuration data in order to deploy services in the proper mission configuration.
Check Software and Firmware Version	This function enables the system to reference Software and Firmware version data.
Collect Health and Status Data	This function receives and processes health and status data from Level 1 and Level 2 MPUs. The function also infers or calculates additional information beyond what is directly measured.
Consolidate Fault Data	This function consolidates duplicate fault data.
Consume Built-In Test Information	This function receives and processes BIT data from Level 2 and Level 3 MPUs.
Deploy Containerized Services	This function deploys services on the appropriate processing hardware.
Facilitate Networking Connectivity	This function facilitates endpoint connections between services.

Name	Documentation
Manage Service Lifecycles	This function provides for the deployment, startup, execution, and shutdown of AMS GRA mission-specific and agnostic services based on the deployment and recovery strategy and utilizing an OCI-compliant Orchestration Engine.
Manage System Faults	This function manages the system faults. It supports the reaction to fault detection of MPUs. This support includes receiving fault data, aggregating disparate fault data, and relaying faults.
Monitor Computing Resource Information	This function tracks the capability and capacity of the computing resource. Information monitored may include system attributes such as the processor cores, volatile and non-volatile memory, and network connection.
Monitor Computing Resources	This function allocates and monitors service usage of computing resources.
Monitor Health and Status	This function monitors the health and status of OMS and AMS Services and hardware elements. The function also allows for collection, processing, and publication of health and status data to the PVI.
Process Health and Status Data	This function processes collected health and status data to determine the overall health of the system.
Provide Execution Framework	This function provides the framework for the execution of mission-specific or mission-agnostic services on the target hardware.
Provide Local Function Information	This function provides a mechanism to allow services to access available Local Function information.
Redeploy Containerized Services	This function redeploys a service on new processing hardware in the case of a hardware failure.
Report Computing Resource Data	This function publishes physical computing information which may include processor, memory, and network bandwidth capability.

Name	Documentation
Respond to Fault Data	This function detects when an AMS GRA mission-Specific/Agnostic Service unintentionally terminates and determines what processing resources are required to relocate and restart the service resources based on the deployment and recovery strategy. This function also reacts to degraded hardware processing resources, typically resulting in redeployment of software services.
Shutdown Containerized Services	This function terminates a deployed containerized service.
Start Containerized Services	This function starts the execution of a deployed containerized service.
Stop Containerized Services	This function stops the execution of a deployed containerized service.
Store Fault Data	This function stores fault data in persistent data storage.
Support Software and Firmware Versioning and Configuration	This function enables MASI to collect and reference stored version and configuration data of Software and Firmware in order to deploy services in the proper version and configuration.
Translate Health and Status Data	This function makes processed health and status data received from the ABB available for use on the ASB.

Table 3.1.1: The MASI Required Functions Outlines Minimum Functionality for a Compliant MASI Implementation. These Functions Must be Present in Order to Achieve Any Compliance Tier.

3.2 MASI Interfaces

Figure 3.2.1 illustrates how MASI may communicate with other elements within the AMS GRA architecture. It is not intended to depict port data flows of the MASI, only potential interfaces to support.

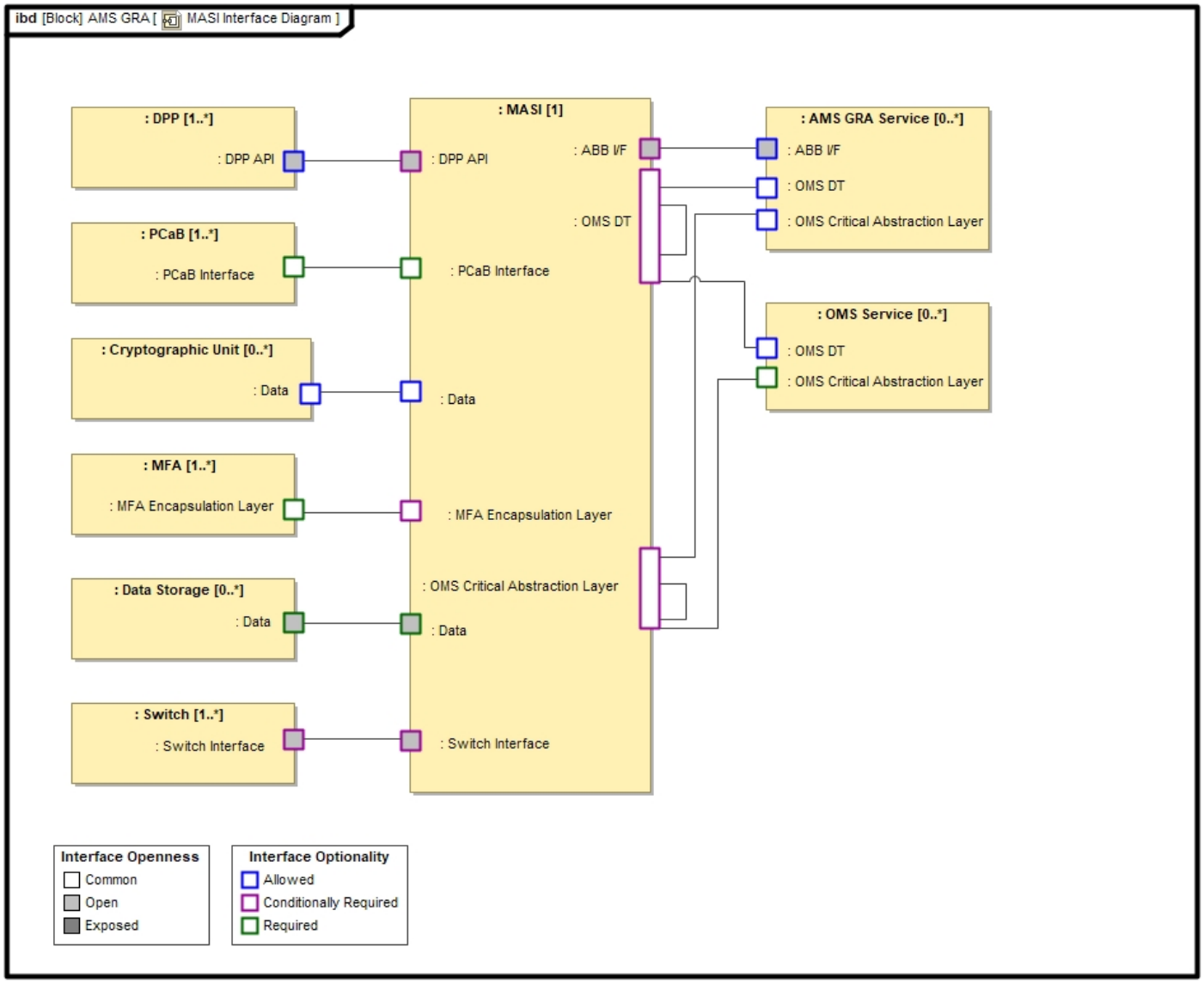


Figure 3.2.1: The MASI Interface Diagram Details How the MASI Interfaces Connect to Other Architectural Elements of the AMS GRA.

3.2.1 MASI Management Interface

The purpose of the MASI Management Interface is to flexibly satisfy common health and status reporting data flows for DPP, Switch, Cryptographic Unit, CDS, and Data Storage hardware MPUs as stipulated in GRA-MPU-088. These include component identification and descriptions, state and fault reporting, and tailoring capabilities that permit growth reporting for MPU-specific status and performance measures.

The MASI Management Interface Application Programming Interface (API) is specified so as to standardize "Monitor Health and Status" functional reporting, primarily through "Fault Information" and "Hardware Health" MASI Data Exchanges. The wide variety of other MASI Data Exchanges, documented in **Section 3.2.2** of the MASI architectural volume, are not currently subject to a common API.

AMS GRA interface implementations are not prescriptive, and the MASI can choose any AMS-GRA-compliant option for the mission system at-hand to implement the MASI Management Interface. For instance, an MSI may choose to implement status reporting for the API by leveraging Universal Command and Control Interface (UCI) across the ASB where feasible. The MASI Management Interface is defined by the data exchanges described in **Table 3.2.2**.

The common portion of the API is colored green in **Table 3.2.1**. (Optional details for finer-grain status follows.)

Data	Multiplicity	Notes
Component Identification (ID)	1	ID designating top-level granularity for hardware MPU
Component Universally Unique Identifier (UUID)	1	
Component text description	1	
Component State	1	Enumeration defined by MSI in conjuction with MPU suppliers
Data Timestamp	1	
Diagnostic test ID	1	Includes Built-In Test (BIT)
Diagnostic test description	1	Includes BIT
Specific Status	0..∞ Optional sub-component status outlined below	Provides for sub-component details as desired

Data	Multiplicity	Notes
Component Element UUID	1	ID designating sub-component granularity for hardware MPU
Component Element Text Description	1	
Component Element State	1	Enumeration defined by MSI in conjuction with MPU suppliers
Component Element State Text Description	1	
Component Detailed Status	0..∞ Provided for MPU-specific status using status name-value-text triples as desired	Defined by MPU supplier
Detailed Status Name	(see Component Detailed Status)	Defined by MPU supplier
Detailed Status Text Description	(see Component Detailed Status)	Defined by MPU supplier
Detailed Status Text Description	(see Component Detailed Status)	

*Table 3.2.1: As MASI matures and additional use cases are defined, the MASI Management Interface API may grow to encompass more of the data listed in **Table 3.2.2**.*

3.2.2 MASI Data Exchanges

Table 3.2.2 identifies the MASI data exchanges depicted in **Figure 3.2.1**. The third column of the table indicates whether the data exchange is required, conditionally required or allowed. To determine whether a given MPU or infrastructure is AMS GRA compliant, the implementor must provide all the required and the necessary conditional interfaces. The three interface categories are defined as follows:

- Required data exchange must be implemented.
- Conditionally required data exchanges must be implemented when specific conditions are satisfied.
- Allowed data exchanges are optional to implement.

Name	Documentation	Optionality	Notes
Compute Information	MASI uses this interface to receive computing performance data such as memory, CPU Core, and OS Disk utilization information from MPUs.	Required	May be included in the MASI Management Interface for Digital Payload Processor (DPP) MPUs
Critical Program Information	MASI uses this interface to receive platform CPI.	Conditionally Required	
Cyber Information	MASI uses this interface to receive and respond to requests of the AMS GRA Cyber Resiliency (CR) Framework and cyber services external to MASI.	Required	May be included in the MASI Management Interface for Cryptographic Unit MPUs
Disk Information	MASI uses this interface to provide networked storage information to services, allowing them to access Data Storage MPUs.	Conditionally Required	
Endpoint Information	MASI uses this interface to receive and collect endpoints and respond to service endpoint requests from non-MASI services.	Required	
Fault Information	MASI uses this interface to receive MPU fault status, including fault severity and fault code.	Required	Included in the MASI Management Interface for hardware MPUs
Hardware Endpoint Information	MASI uses this interface to receive responses to query and store Level 2 MPU endpoints.	Allowed	
Hardware Health	MASI uses this interface to receive information for the current health of the hardware.	Required	Included in the MASI Management Interface for hardware MPUs
Logging Function	MASI uses this interface to receive logging data.	Required	
Memory Availability	MASI uses this interface to receive and respond to queries for availability of volatile memory.	Required	May be included in the MASI Management Interface for Data Storage MPUs
Networking Information	MASI uses this interface to receive and collect the current bandwidth, latency status, and availability of the network.	Allowed	
Orchestration Engine Information	MASI uses this interface to receive and collect orchestration status and configuration data.	Allowed	

Name	Documentation	Optionality	Notes
Performance Data	MASI uses this interface to receive performance data from MPUs and Skills.	Allowed	May be included in the MASI Management Interface for hardware MPUs
Platform State	MASI uses this interface to receive platform state changes that will trigger MASI services to initiate lifecycle changes.	Conditionally Required	
Processing Core Availability	MASI uses this interface to receive and respond to CPU processing core availability requests.	Required	
Security Configuration	MASI uses this interface to receive data files indicating what security configuration the system should be deployed to match.	Allowed	
Security Information	MASI uses this interface to receive security information from the Mission System or the platform.	Allowed	May be included in the MASI Management Interface for Cross Domain Solution (CDS) MPUs
Service Data	MASI uses this interface to receive and respond to query for processed service data.	Allowed	
Service Health	This interface is used to report the current MASI service performance to the Mission System.	Required	
Service Health and Status	MASI uses this interface to receive and collect the current performance and state of services.	Required	
Service Lifecycle Information	This interface is used to receive information in tasking commands from the Mission System and respond with commands to deploy, terminate, and shutdown services.	Required	
Service Status	MASI uses this interface to receive the reported current status information of the service.	Required	
Software Endpoint Information	MASI uses this interface to receive and respond to queries and store Level 1 MPU endpoints.	Required	

Name	Documentation	Optionality	Notes
Storage Information	MASI uses this interface to receive and respond to queries for Data Storage capacity, persistent storage availability, Data Storage availability, status, and data to be stored.	Required	May be included in the MASI Management Interface for Data Storage MPUs
Stored Data Requests	MASI uses this interface to receive and respond to queries for data in persistent storage.	Conditionally Required	
Subsystem Status	MASI uses this interface to report the aggregated status of Skills and hardware to the MPS.	Required	
System Data	MASI uses this interface to publish and receive and respond to queries for the overarching system status to the Mission System.	Required	
Time Interface	MASI services receive time from the Mission System.	Required	

Table 3.2.2: The Procuring Authority, MPU Provider, and MSI Use this Table to Identify the Data Exchanges to Implement for a Compliant MPU or Infrastructure.

4.0 Compliance Requirements

The requirements in **Table 4.0.1** are the minimum set that any MASI implementation would need to satisfy in order to be AMS GRA-compliant. **Table 4.0.1** includes requirements as part of the Security Framework that need to be instantiated by a MASI implementation.

See the Mission System Volume for the requirements regarding the referenced CR Monitor Service and CR Manager Service.

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MASI-001	A MASI shall be able to start mission-specific and Mission-Agnostic Services.	Tier 1	Demonstration
GRA-MASI-002	A MASI shall be able to stop mission-specific and Mission-Agnostic Services.	Tier 1	Demonstration
GRA-MASI-003	A MASI shall monitor the Management Interface for reported health and status of AMS GRA-compliant hardware.	Tier 1	Analysis
GRA-MASI-006	A MASI shall monitor fault information reported by AMS GRA-compliant MPUs.	Tier 1	Inspection
GRA-MASI-008	A MASI shall consume all available BIT information reported by AMS GRA-compliant hardware.	Tier 1	Analysis
GRA-MASI-010	A MASI shall provide a mechanism to aggregate disparate fault, health, and status data.	Tier 2	Inspection
GRA-MASI-011	A MASI shall provide a mechanism to store fault, health, and status data.	Tier 2	Inspection
GRA-MASI-012	A MASI shall be able to respond to fault information reported by AMS GRA-compliant MPUs.	Tier 2	Analysis
GRA-MASI-013	A MASI shall provide an interface to access fault, health, and status data.	Tier 2	Inspection
GRA-MASI-015	A MASI shall monitor the health and status reporting of all deployed AMS GRA and OMS Services.	Tier 1	Inspection
GRA-MASI-016	A MASI shall monitor compute resource information (e.g., utilization, availability, capability, bandwidth, flash, RAM).	Tier 2	Inspection
GRA-MASI-017	A MASI shall provide a mechanism to allow services to access all available software and firmware configuration status and versioning information for software and firmware stored and deployed on the AMS GRA Mission System.	Tier 3	Demonstration
GRA-MASI-020	A MASI shall provide a mechanism to consolidate duplicate fault data.	Tier 2	Inspection
GRA-MASI-021	A MASI shall include an OCI-compliant Orchestration Engine.	Tier 2	Inspection
GRA-MASI-022	A MASI shall deploy OCI-compliant containerized software utilizing a container framework.	Tier 1	Demonstration
GRA-MASI-023	A MASI shall provide access to compute resource information.	Tier 2	Inspection
GRA-MASI-024	A MASI shall facilitate endpoint connections between services.	Tier 1	Demonstration
GRA-MASI-025	A MASI shall provide a mechanism to allow services to access Local Function information.	Tier 2	Inspection
GRA-MASI-027	A MASI shall register the services available for deployment.	Tier 1	
GRA-MASI-028	A MASI shall report the state of MASI	Tier 1	
GRA-MASI-029	A MASI shall support deployment of a minimum of two instances of the same Service.	Tier 1	Test
GRA-MASI-030	A MASI shall isolate AMS GRA and OMS Service software faults from other software MPUs.	Tier 1	Test

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MASI-031	A MASI shall assign an MPUs' deployment targets and resources in accordance with that MPUs's delivered service.yml file.	Tier 1	Test
GRA-MASI-032	A MASI shall provide an implementation of the AMS Common Transport API.	Tier 1	
GRA-MASI-033	A MASI shall report the status of all SW MPUs available in the system in UCI format.	Tier 2	Test
GRA-MASI-034	A MASI shall report compute resource information in UCI format if requested in UCI format by an MPU within the system.	Tier 2	Test

Table 4.0.1: A MASI's Adherence to the Compliance Requirements Defines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
ASB	Abstract Service Bus
BIT	Built-In Test
CDS	Cross Domain Solution
CPI	Critical Program Information
CPU	Central Processing Unit
CR	Cyber Resiliency
DPP	Digital Payload Processor
GRA	Government Reference Architecture
ID	Identification
IDD	Interface Description Document
MASI	Mission Agnostic Service Infrastructure
MPS	Mission Processing Subsystem
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
OCI	Open Container Initiative
OMS	Open Mission Systems
OS	Operating System
PVI	Pilot Vehicle Interface
RAM	Random Access Memory
SK	Starter Kit
UCI	Universal Command and Control Interface
UUID	Universally Unique Identifier

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the SK continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision did not introduce any significant changes.	July 24, 2025	SK13.0
B	This revision adds **Section 3.2** and updates figures and tables throughout.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the MASI Volume Between Documentation Releases.